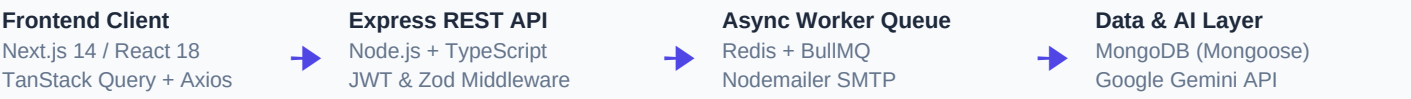


1. System Overview & Architectural Strategy

SkillTrack is engineered as a high-performance, decoupled monorepo web application. The client application is built with **Next.js 14 (App Router)** and **React 18** for dynamic rendering, optimized page loads, and modern UI components. The backend is an **Express.js / Node.js** REST API using **MongoDB** with **Mongoose ODM** for core persistence. High-throughput asynchronous tasks (such as AI generation pipelines and transactional emails) are decoupled via **Redis** and **BullMQ** worker queues. Intelligent features leverage the **Google Gemini API** enriched with content from the **YouTube Data API**.



2. Core Stack Breakdown (Frontend vs Backend)

FRONTEND LAYER

- **Next.js 14 & React 18:** App router framework with Server-Side Rendering (SSR) & Client Components.
- **TypeScript (v5.4):** End-to-end static typing across pages, components, and API client interfaces.
- **Tailwind CSS 3.4 & Radix UI:** Custom utility-first design system with accessible primitives (`@radix-ui/react-*`).
- **Framer Motion & Lucide React:** Fluid micro-animations, transitions, and modern vector iconography.
- **TanStack React Query (v5) & Axios:** Client-side asynchronous state caching and REST API integration.
- **React Hook Form & Zod:** High-performance form state management with strict schema validation.
- **Recharts & Sonner:** Interactive analytics dashboard charts and dynamic toast feedback.
- **@react-oauth/google:** Google OAuth 2.0 single sign-on integration.

BACKEND LAYER

- **Node.js & Express.js (v4):** Modular REST API server with TypeScript runtime tooling (`ts-node-dev`).
- **MongoDB & Mongoose ODM (v8):** Document database for users, roles, roadmaps, and college data.
- **Redis & BullMQ (v6):** High-performance in-memory caching and distributed async background task worker queues.
- **JWT & bcryptjs & Google Auth:** Secure token authentication, password hashing, and OAuth ID token verification.
- **Zod Schema Middleware:** Strict runtime validation of HTTP request bodies, parameters, and query strings.
- **Multer File Engine:** Parsing multipart forms for career outcome verification proof uploads.
- **Nodemailer & SMTP:** Transactional email dispatch system integrated into BullMQ queues.
- **Morgan & CORS:** Logging middleware and cross-origin resource sharing controls.

3. AI Core Engine & External Services

■ Google Gemini AI Engine  
(`@google/generative-ai`)

- **Adaptive Question Generation:** Generates dynamically scaled assessment questions based on student performance.
- **AI Skill Gap Analysis:** Evaluates user competencies against market career roles.
- **Personalized Roadmaps:** Creates custom milestone learning paths.
- **Placement Readiness Scoring:** Calculates real-time readiness index.

■ External APIs & Services

- **YouTube Data API v3:** Fetches curated learning videos for specific roadmap skills.
- **Google OAuth 2.0:** Provides seamless single sign-on authentication.
- **SMTP Email Gateway:** Transactional email notifications via BullMQ background queue.
- **Local File Storage:** Stores uploaded evidence files for outcome verification.

4. Feature Module & Technology Mapping Matrix

| Feature Module        | Frontend Stack                | Backend Controller / Service               | Data Layer / Queue   |
|-----------------------|-------------------------------|--------------------------------------------|----------------------|
| Adaptive Testing      | React State, Radix Dialog     | `aiQuestion.service.ts`                    | MongoDB Assessments  |
| Skill Gap & Readiness | Recharts Radar & Bar Charts   | `aiSkillGap.service.ts` /<br>`aiReadiness` | MongoDB User Skills  |
| Learning Roadmaps     | Framer Motion Timeline UI     | `aiRoadmap.service.ts` + YouTube API       | MongoDB Roadmaps     |
| Async Email Dispatch  | Sonner Toast Feedback         | `email.service.ts` (Nodemailer)            | Redis + BullMQ Queue |
| Outcome Verification  | React Hook Form + File Upload | `careerOutcomeVerification.service.ts`     | Multer + MongoDB     |